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Biochemical and Biophysical Research Communications 334 (2005) 674–680
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Kinetics of calmodulin binding to calcineurin

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Received 23 June 2005
Available online 5 July 2005

Abstract

Calcineurin (CaN) binds Ca^{2+}-saturated calmodulin (CaM) with relatively high affinity; however, an accurate steady-state K_d value has not been determined. In this report, we describe, using steady-state and stopped-flow fluorescence techniques, the rates of association and dissociation of Ca^{2+}-saturated CaM from CaN heterodimer (CaNA/CaNB) and CaNA only. The rate of Ca^{2+}/CaM association was determined to be 4.6×10^7 M^{-1} s^{-1}. The rate of Ca^{2+}/CaM dissociation from CaN was slower than previously reported and was approximately 0.0012 s^{-1}. In preparations of CaNA alone (no regulatory CaNB subunit), the dissociation rate was slowed further to 0.00026 s^{-1}. From these data we calculate a K_d for binding of Ca^{2+}-saturated CaM to CaN of 28 pM. This K_d is significantly lower than previously reported estimates of ~1 nM and indicates that CaN is one of the highest affinity CaM-binding proteins identified to date.
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Keywords: Calmodulin; Calcineurin; Calmodulin-binding proteins; Steady-state fluorescence; Stopped-flow fluorescence; Kinetics

Calmodulin binds to a wide variety of proteins with diverse and essential downstream cellular effects [1–3]. Some preferentially bind to CaM in the absence of Ca^{2+} while others favor binding in the presence of Ca^{2+}. In the context of the intracellular environment, these various targets must compete for the binding of CaM as Ca^{2+} concentrations rise and fall. These processes must be well coordinated to avoid futile cycles that might occur, for example, by the simultaneous activation of CaM-dependent phosphodiesterases and CaM-dependent adenylyl cyclases. Phosphorylation/dephosphorylation cycles mediated by CaM-dependent protein kinases and CaM-dependent protein phosphatases require similar orchestration. One key factor governing the dynamics of these cycles is the kinetics of CaM binding to the various targets. If the association rate of a particular enzyme is slow it will not effectively compete with other enzymes during the rising phase of a Ca^{2+} pulse. On the other hand, if the dissociation rate is slow once bound, the enzyme will retain CaM for a significant time during the falling phase of a Ca^{2+}-pulse, both making CaM unavailable for binding to other targets and producing an active state of the enzyme that potentially outlives the Ca^{2+} pulse. Determining the rates of association and dissociation from CaM-dependent enzymes can provide a qualitative sense for such effects, but the values themselves are imperative for accurate quantitative modeling of cellular signaling.

Calcineurin (CaN) is the only Ca^{2+}/CaM controlled phosphatase found in cells [4]. It is a phospho-Ser/Thr directed phosphatase with relatively broad substrate specificity. In its native state, CaN exists as a heterodimer: CaNA is the catalytic subunit and CaNB is the regulatory subunit. The activity of CaN is suppressed through an autoregulatory mechanism where the C-terminal domain of CaNA binds to and inhibits the catalytic site. The binding of Ca^{2+}-saturated CaM relieves this inhibition, activating the phosphatase. Like CaM, with which it shares 35% sequence identity, CaNB binds four Ca^{2+} ions via the EF hand motif [4,5]. CaNB is tightly associated with CaNA and can be extracted only

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0006-291X/$ - see front matter © 2005 Elsevier Inc. All rights reserved.
doi:10.1016/j.bbrc.2005.06.152

under denaturing conditions or at extremely low Ca^{2+} concentrations [6]. The binding of CaNB to CaNA is essential for maximal activation by Ca^{2+}/CaM [7].

The present study was undertaken to provide kinetic values for the rates of association and dissociation of CaM from CaN. Previous studies have determined approximate K_ds for CaM-binding [8], and one study determined a rate for Ca^{2+}/CaM dissociation from CaN [9]. To complete the picture of CaM-binding kinetics, we determined in the present study the association and dissociation rates of Ca^{2+}/CaM-binding to CaN. We also measured the rate of Ca^{2+}/CaM dissociation from the CaNA subunit in the absence of CaNB. We further determined the rate of CaM dissociation from both CaN and from CaNA only when Ca^{2+} was removed from the complexes. Surprisingly, our data show that the dissociation rate of Ca^{2+}/CaM from CaN is two orders of magnitude slower than that previously reported. Based on the measured rates, a K_d for binding of Ca^{2+}/CaM to CaN was calculated. That value is approximately 40-fold lower than previously estimated values [8]. Thus, the affinity of CaN for Ca^{2+}/CaM is one of the highest (numerically lowest) in the family of CaM-dependent enzymes. This suggests that CaN would compete very effectively for CaM in the intracellular setting.

Experimental methods

Calmodulin expression, purification, and labeling. A codon optimized wild-type CaM, originally from sea urchin, was mutated to contain a single cysteine residue at amino acid 75; CaM(C75). Wild-type (wt) and CaM(C75) both were expressed in *Escherichia coli* and purified as described [10]. CaM(C75) was labeled with Acrylodan (ACR) exactly as described in [11].

CaN expression and purification. Stocks of baculovirus engineered to express the A and B subunits of rat CaNα were obtained from Dr. Brian Perrino and expressed in Sf21 insect cells [7,9]. Our standard protocol was to infect cells in 1 L spinner flasks simultaneously with independent stocks of virus expressing the A or B subunit. In some experiments, only the virus expressing CaNA was used to infect cultures. In other experiments, the cells were infected with virus expressing the B subunit 24 h before infecting with the A subunit in attempts to optimize the saturation of A subunit with B subunit. Enzyme purified from cells infected with this latter protocol or our standard simultaneous infection did not exhibit an identifiable difference in the ratio of A to B subunits when analyzed by SDS–PAGE.

To purify CaN, cell pellets were lysed in ice cold lysis buffer (50 mM Tris–HCl, pH 7.5, 6 mM $MgSO_4$, 1 mM EGTA, and 0.5 mM DTT containing 5 μg/ml leupeptin hemisulfate, 0.1 mM phenylmethylsulfonyl fluoride, and 0.02 mg/ml of soybean trypsin inhibitor) and then sonicated briefly. The lysate was then centrifuged at 10,000*g* for 30 min and the supernatant was then brought to a final concentration of 45% ammonium sulfate. After 10 min mixing at 4 °C, the precipitated protein was collected by centrifugation at 7000*g*. The pellets were then resuspended in CaM–Sepharose start buffer (20 mM Hepes, pH 7.4, 0.1 M NaCl, 2 mM $CaCl_2$, and 10% glycerol). The protein sample was then loaded onto a 5 ml CaM–Sepharose column (~10 mg/ml CaM bound to the gel matrix) at a flow rate of ~1 ml/min. The column was then washed with 10 ml of each of the following: start buffer, start buffer including 1 M NaCl, and start buffer again. CaN was eluted from the column using elution buffer (20 mM Hepes, pH 7.4, 0.5 M NaCl, 2 mM EGTA, and 10% glycerol) and 1 ml fractions were collected. The A_{277} was determined for each fraction and the preliminary concentration values were calculated using ε_{277} of 1.0 = 9.3 mg/ml [8]. Peak fractions were dialyzed overnight at 4 °C into 20 mM Hepes, pH 7.4, 100 mM KCl with 10% glycerol. For on-rate measurements, which required high protein concentrations, the peak fractions were dialyzed into the same buffer without glycerol and used within two days without freezing. The proteins were ~95% pure as shown by the Coomassie stained SDS–PAGE.

CaN activity measurements. Each purified CaN preparation was assessed for activity using spectrophotometric assays based on either pNPP as substrate or phosphorylated peptide from the regulatory subunit of protein kinase A (RII-P) as described by Perrino et al. [9], with some modifications. The hydrolysis of pNPP to pNP was monitored by measuring A_{405} in a microtiter plate based assay and all assays were accomplished with 80 mM pNPP (final concentration). Removal of phosphate from the RII-P substrate was assessed with the BioMol assay kit exactly as described by the manufacturer with the exception that color generation was quantified by monitoring for A_{595} instead of A_{620}. A phosphate standard curve was run with every BioMol assay to calculate specific activity of the enzyme preparations. For both assays, the enzyme reaction buffer was 50 mM Tris, pH 8.6, 20 mM $MgCl_2$, 1 mM dithiothreitol, 0.1 mg/ml bovine serum albumin, and either 0.1 mM $CaCl_2$ or 2 mM EGTA as described [12]. Neither assay was particularly sensitive and required relatively high concentrations of CaN to produce good signal over background. Typically, the assays were accomplished with ~25–50 nM CaN in each reaction (final concentration). At these concentrations, the activation curve reflects CaM titering the amount of CaN present, and the experiment is best interpreted as a dose–response curve. In this regard, the K_{act50} is entirely dependent on the CaN concentration in the assay. In addition, the biological activity of $CaM(C75)_{ACR}$ was assessed by producing a titration curve of the protein, along with wt CaM, in its ability to activate CaN. The K_{act} was not significantly different and the V_{max} for $CaM(C75)_{ACR}$ was only slight reduced (25%) relative to wt CaM (data not shown). This indicates that $CaM(C75)_{ACR}$ is a reasonable reagent to test the binding kinetics of CaM to CaN.

Rate constant measurements. The dissociation rate of Ca^{2+}/CaM from CaN was determined using steady-state fluorimetry in a PTI QuantaMaster spectrofluorimeter. $CaM(C75)_{ACR}$ (2.5–200 nM) was used as the fluorescent reporter that exhibited excitation and emission maxima of 375 and 456 nm, respectively. The slit widths were typically set at 2 nm for excitation and 7 nm for emission. All additions were made manually into a 2 ml quartz cuvette with constant stirring. The reaction buffer contained 25 mM Mops, pH 7.0, 150 mM KCl, 100 μM EGTA, and 0.1 mg/ml of bovine serum albumin. Other reagents were diluted into start buffer without EGTA. The protocol was to collect 30 s of baseline data, add $CaCl_2$ (typically 5 mM final) and collect an additional 30 s of data, add a saturating concentration of CaN and collect an additional 30 s of data, and finally add a 50-fold excess of unlabeled CaM (relative to the amount of labeled CaM used in the reaction) and monitor the decrease in fluorescence intensity over time. The rate of dissociation was determined by fitting the data with a single exponential decay function in Origin (Origin Lab). Ideally, the final fluorescent value after the addition of excess CaM would have returned to the initial fluorescence value before addition of CaN, representing 100% exchange of the labeled CaM from CaN. We found that this percent recovery varied from prep to prep between 70% and 95%. These differences could not be ascribed to observable differences in the ratio of CaNB to CaNA in the purified enzyme preparations.

Ca^{2+}/CaM association rates. Association rates were measured using an Applied Photophysics (Leatherhead, UK) model SV.17 MV sequential stopped-flow spectrofluorimeter with a dead time of 1.7 ms. Excitation was from a 150 W Hg/Xe lamp at 365 nm with a 5 nm slit width and emitted light was collected through a long-pass cut off filter

(Orion). Association rates were measured by rapidly mixing equal volumes of 50 nM $CaM(C75)_{ACR}$ in 20 mM Mops, pH 7.0, 150 mM KCl, and 5 mM $CaCl_2$ with serial dilutions of CaN (from 2 μM to 125 nM to create pseudo-first order conditions) and measuring the fluorescence increase due to binding of CaM to CaN. Data from 3 to 5 injections were collected and averaged at each concentration of CaN and then fit with a single exponential function. The final on-rate constant was determined by the slope of a plot of CaN concentration versus the measured rate of association.

Kinetics of CaM dissociation induced by EGTA. To determine the rate of Ca^{2+}/CaM dissociation from CaN by removal of Ca^{2+}, stopped-flow experiments were performed exactly as described above. In this case, $CaM(C75)_{ACR}$ pre-associated with CaN in standard Ca^{2+}-containing reaction buffer and then was mixed with a solution of 10 mM EGTA in reaction buffer. The final Ca^{2+} concentration under these experimental conditions was calculated to be <5 nM [13]. Dissociation was detected by monitoring the rate of fluorescence decrease. Five to six injections were averaged to produce the final traces shown and all experiments were reproduced three independent times. It should be noted that the rates of fluorescence decrease in this circumstance reflect the rates of Ca^{2+} dissociation from $CaM(C75)_{ACR}$, and also of $CaM(C75)_{ACR}$ dissociating from CaN; both produce a fluorescence decrease.

Results

Calcineurin was purified to near-homogeneity from baculovirus-infected Sf21 cells using CaM–Sepharose chromatography. The purified protein is a heterodimer of two subunits, CaNA and CaNB, of approximately 60 and 18 kDa, respectively (Fig. 1A, lane c). In these preparations the CaNB subunit migrates on SDS–PAGE as two molecular weight species and the upper molecular weight band is consistent with a non-myristoylated version of CaNB as described previously [14]. CaN lacking the CaNB subunit was also expressed and purified in an identical fashion. There was no evidence of the CaNB subunit in these preparations (Fig. 1A, lane b). The activity of these preparations was characterized previously and we verified the Ca^{2+}/CaM-dependent activation profile using as substrate a phosphorylated synthetic peptide mimicking a site on the regulatory subunit of PKA (RII-P). Using the RII-P substrate a K_{act} of approximately 14 ± 4 nM was determined, and the specific activity of the preparations was 0.32 μmol Pi/mg/min (Fig. 1B). Similar activity profiles were also produced using pNPP as substrate although the K_{act} was determined to be approximately 3.1 ± 1 nM CaM. As reported previously, CaNA in the absence of CaNB expresses no detectable activity under the assay conditions here (Fig. 1B).

The kinetics of CaM association and dissociation from CaN in the presence of Ca^{2+} were determined using steady-state and stopped-flow fluorescence techniques. $CaM(C75)_{ACR}$ was shown previously to be an effective tool for monitoring the association and dissociation of CaM from CaM-kinase II [10,11]. To determine the usefulness of $CaM(C75)_{ACR}$ in examining CaM and CaN interactions, we first determined the spectral characteristics of $CaM(C75)_{ACR}$ binding to CaN. As shown in Fig. 2, there was no intensity or spectral shift in the peak emission of $CaM(C75)_{ACR}$ when excess CaN was added to the cuvette. These reactions were started in a buffer containing 100 μM EGTA and indicated that little interaction occurred between CaN and CaM in the absence of Ca^{2+}. Upon the addition of Ca^{2+}, there was a large (almost 200%) increase in fluorescence intensity and a distinct blue shift in the emission maxima, from 495 to 456 nm. A blue shift in emission maximum is consistent with the Acrylodan moiety moving into a more apolar environment. Upon removing Ca^{2+} with an excess of EGTA, the emission spectra returned to the pre-Ca^{2+} intensity and emission maximum. These

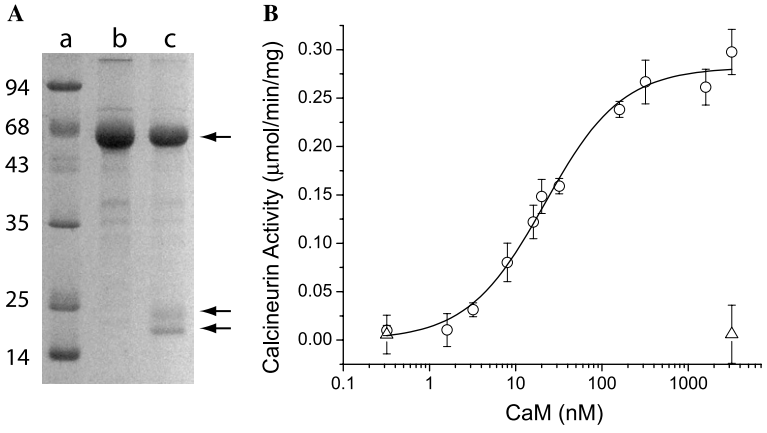


Fig. 1. Characterization of baculovirus-expressed CaN. CaN or CaNA only was purified from baculovirus-infected cells as described in methodology. (A) Five micrograms of each preparation was separated by SDS–PAGE and stained with Coomassie blue. Lane b is CaNA only and lane c is CaN (CaNA and CaNB). The arrows indicate the relative positions of CaNA and CaNB. CaNB migrates as a doublet, likely due to differential myristoylation. (B) CaM activation curve for CaN. Increasing concentrations of CaM were added to a fixed amount of CaN (50 ng) in standard reaction buffer and the dephosphorylation of RII-P was assessed as described in methodology. The data were fit to the Hill equation. Data are means ±SD. CaNA only was assessed at the maximal CaM concentration (open triangle) and there was no significant activity detected.

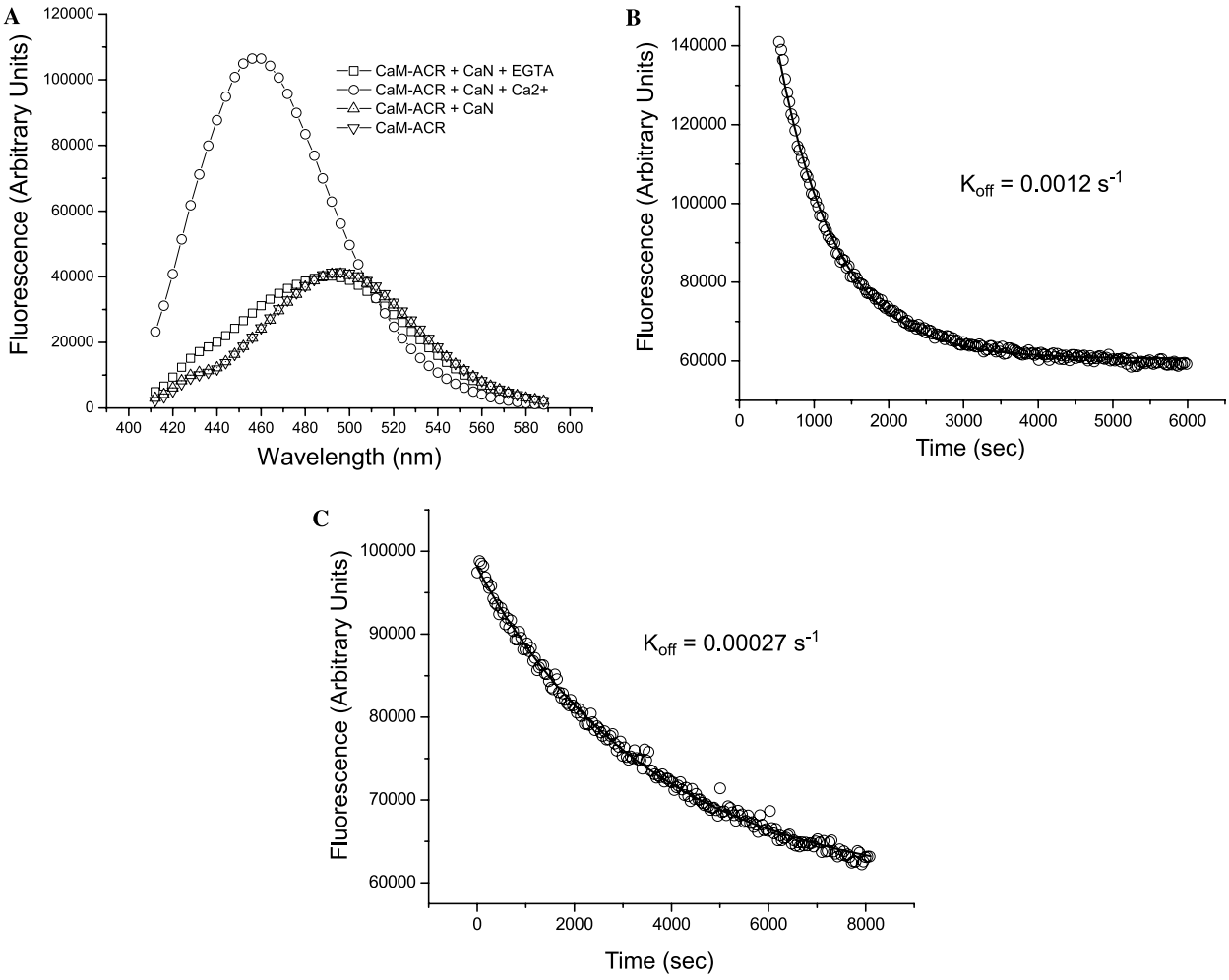


Fig. 2. Steady-state fluorescence of CaM-binding to CaN. (A) Characterization of emission spectra from Ca^{2+}-binding and CaN binding to $CaM(C75)_{ACR}$. $CaM(C75)_{ACR}$ (60 nM) was added to 1 ml of 20 mM Mops, pH 7.0, 150 mM KCl, 0.1 mM EGTA, and 1 mg/ml BSA and emission scans were obtained between 400 and 590 nm (inverted triangles). The excitation wavelength was set at 375 nm and the slit widths were set to 2 and 7 nm for excitation and emission, respectively. CaN was added (100 nM final) and a second emission scan was obtained (triangles). Note no significant increase was detected in fluorescence intensity. $CaCl_2$ (0.5 mM final) was then added and emission scan obtained (circles). A significant increase and blue shift was observed in the peak of fluorescence intensity. Finally, EGTA was added (10 mM final) and a scan obtained (squares). Note the fluorescence intensity returns to a similar level to that before Ca^{2+} was added. There was a slight blue shift in the peak of fluorescence intensity following addition of EGTA. (B) Measurement of Ca^{2+}/CaM dissociation rate by steady-state kinetics. The reaction conditions were similar to those in (A), although the reaction volume was increased to 2 ml and the solution was mixed constantly with a small stir bar. CaN and Ca^{2+} were added and after a stable baseline was achieved, a 50-fold excess of Ca^{2+}/CaM (3 μM final) was added to the cuvette. The decreasing data shown are due to the decrease in fluorescence as Ca^{2+}/$CaM(C75)_{ACR}$ dissociates from CaN. The solid line represents a single exponential fit to the data. (C) Under identical reactions to those described in (B), the rate of dissociation of CaM from CaNA only was determined. The data were also fit with a single exponential function.

data indicated that $CaM(C75)_{ACR}$ was an appropriate tool for measuring CaM-binding to CaN. Furthermore, we determined that $CaM(C75)_{ACR}$ was effective at activating CaN when a titration of CaM and $CaM(C75)_{ACR}$ against CaN using pNPP as the substrate revealed little (<10%) difference in either K_{act} or in V_{max} (data not shown).

Steady-state fluorimetry was used to determine the rate of Ca^{2+}/CaM dissociation from CaN as described under Experimental methods. When a 50-fold excess of Ca^{2+}/CaM is added to a preformed complex of Ca^{2+}/$CaM(C75)_{ACR}$/CaN, a slow, steady decrease in fluorescence is detected as the labeled CaM dissociates from CaN and is replaced by the unlabeled CaM. This decrease in intensity could be fit with a single exponential decay function and revealed a dissociation rate of 0.0012 s^{-1}. This experiment was repeated many times with four different preparations of CaN. A complete exchange was not always detected in these experiments, suggesting that a fraction of the $CaM(C75)_{ACR}$ was binding to CaN in a manner that did not exchange under the time frame of these experiments. The non-exchangeable fraction varied from preparation to preparation between 5% and 30%, but the rates of

Ca^{2+}/CaM dissociation were relatively consistent and varied between 0.0011 and 0.0029 s^{-1}. A K_{off} of $0.0013 \pm 0.0005\ s^{-1}$ (mean ± SD, $n = 14$) was derived from the data in Fig. 2. We considered whether the variability in the non-exchangeable fraction might be due to differences in the amount of CaNB present in the preparations, leaving a proportion of the CaNA subunits lacking CaNB. This did not appear to be the case as there was no detectable difference in the CaNA/CaNB ratio when each preparation was analyzed by SDS–PAGE (data not shown). We also determined the rate of Ca^{2+}/CaM dissociation from preparations composed of only the CaNA subunit. Interestingly, the Ca^{2+}/CaM dissociation rate was slower by almost an order of magnitude in CaNA only preparations. The K_{off} for Ca^{2+}/CaM dissociation from CaNA only was $2.6 \times 10^{-4} \pm 5 \times 10^{-5}\ s^{-1}$ (mean ± SD, $n = 4$).

The association kinetics of CaM binding to CaN was determined with stopped-flow fluorescence. Experimentally, decreasing concentrations of CaN were rapidly mixed with $Ca^{2+}/CaM(C75)_{ACR}$ and the rate of fluorescence increase was measured. Representative traces are shown in Fig. 3 for only three of the five concentrations of CaN used in this experiment for clarity of presentation. The single exponential fits are shown overtop of the data. The inset shows a plot of the rate of association determined at each concentration of CaN along with a linear regression of the data. The slope of this line reveals an association rate constant of $4.8 \times 10^7\ M^{-1}\ s^{-1}$. This experiment was reproduced five separate times with two different CaN preparations that revealed an average association rate of $4.6 \times 10^7\ M^{-1}\ s^{-1} \pm 0.4$ ($n = 5$, ±SD).

Finally, the rates of $Ca^{2+}/CaM(C75)_{ACR}$ dissociation from CaN and CaNA induced by chelating Ca^{2+} were determined using stopped-flow kinetics. CaN or CaNA was pre-complexed with $CaM(C75)_{ACR}$ in the presence of Ca^{2+} and the rate of dissociation of CaM was assessed by rapid mixing with 10 mM EGTA. Representative traces of this data are shown in Fig. 4. The data with CaN were best fit with a two component model with rates of 2.0 ± 0.3 and $0.2 \pm 0.04\ s^{-1}$. However, the amplitude of the latter component was only 6% of the total fluorescence decrease indicating that the majority of dissociation occurred with a rate of 2.0 s^{-1}. The data for dissociation of $Ca^{2+}/CaM(C75)_{ACR}$ from CaNA could be reasonably well fit with a single component

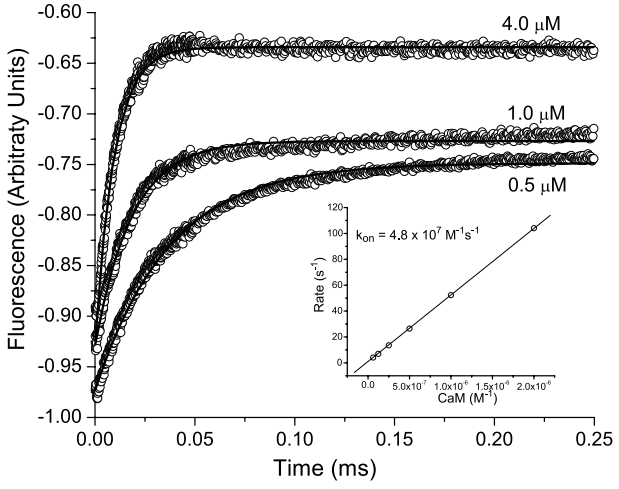


Fig. 3. Measurement of CaM association rate constant for CaN using stopped-flow kinetics. $CaM(C75)_{ACR}$ (100 nM) in 25 mM Mops, pH 7.0, 150 mM KCl, 0.1 mg/ml BSA, and 0.5 mM $CaCl_2$ was rapidly mixed with different concentrations of CaN at 22 °C. Excitation was at 365 nm, and emission was monitored using a 470-nm cutoff filter. Each curve represents the average of five reactions, and for ease of visualization only the data from 0.5, 1, and 4 μM CaN are shown. The solid lines indicate fitting to a single exponential equation. The inset shows a linear relationship between the concentration of CaN and the rate constants as determined at each CaN concentration.

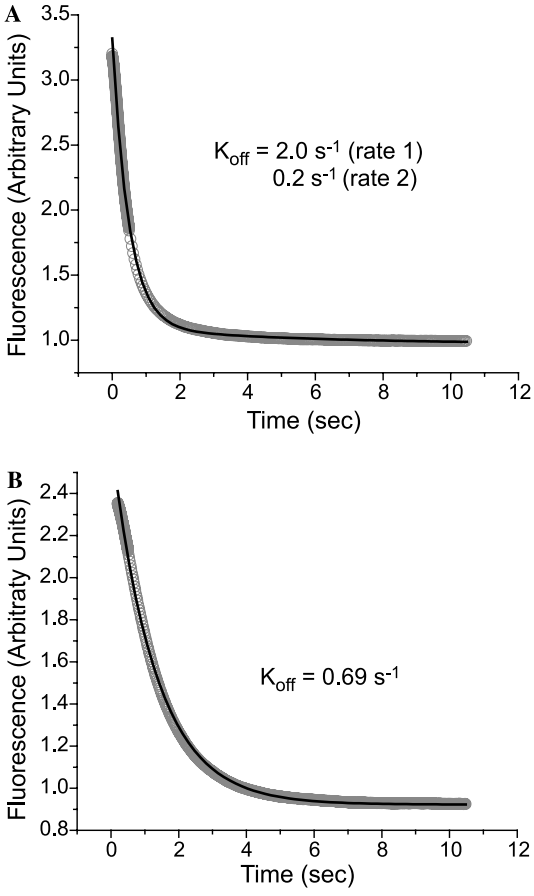


Fig. 4. Measurement using stopped-flow kinetics of the CaM dissociation rate from CaN following chelation of Ca^{2+}. $CaM(C75)_{ACR}$ (120 nM) in 25 mM Mops, pH 7.0, 150 mM KCl, 0.1 mg/ml BSA, 0.5 mM $CaCl_2$, and 200 nM CaN (A) or CaNA only (B) was rapidly mixed with 10 mM EGTA (final) at 22 °C. Excitation was at 365 nm, and emission was monitored using a 470-nm cutoff filter. The data from CaN (A) were fit with a two exponential model and the calculated fits are shown. The data from CaNA only (B) could be fit well with a single exponential equation and the calculated rate is shown. The final free Ca^{2+} under these reaction conditions was <5 nM.

model and the rate determined was $0.69 \pm 0.15\ s^{-1}$. These data are partially complicated by the fact that $CaM(C75)_{ACR}$ exhibits fluorescence intensity changes from both binding Ca^{2+} and binding to target (see Fig. 2) and we cannot determine which step, Ca^{2+} dissociation from $CaM(C75)_{ACR}$ or $CaM(C75)_{ACR}$ dissociation from CaN or both, in the dissociation process is being addressed. Regardless, the overall rate of $2\ s^{-1}$ is relevant in considering the speed of dissociation and, presumably, inactivation of CaN when Ca^{2+} is removed.

Discussion

In this report, we show that CaN binds Ca^{2+}/CaM with one of the tightest binding affinities measured to date (Table 1). Stopped-flow kinetics revealed that the rate of Ca^{2+}/CaM association was $4 \times 10^7\ M^{-1}\ s^{-1}$ and the rate of dissociation was $\sim 0.0013\ s^{-1}$. From these numbers a K_d of 28 pM was calculated (Table 1). To place these values into context, we compared the rates and K_d for CaM-binding to those measured in our laboratory for CaM-kinase II under nearly identical reaction conditions. CaM-kinase II in its basal state has one of the poorest CaM-binding affinities measured ($\sim$76 nM). However, when it is autophosphorylated, CaM-kinase II is described as "trapping" CaM since the affinity increases to approximately 1.8 pM largely due to a slowing of the Ca^{2+}/CaM dissociation rate. There are only modest differences in the association rates of Ca^{2+}/CaM to CaN or to CaM-kinase II in autophosphorylated or unphosphorylated states. Therefore, differences in dissociation rates dominate differences in the K_d. There are few previous reports of the K_d for binding of Ca^{2+}/CaM to CaN. Using solution binding analogous to our studies, Hubbard and Klee [8] reported that the binding affinity of CaN for Ca^{2+}/CaM was <1 nM . Their solid phase experiments indicated a weaker binding affinity. Based on the extremely tight K_d determined from the present data, it would be difficult to develop steady-state assays with enough sensitivity to accurately measure the K_d for binding.

Table 1
Kinetic constants for the interaction of CaM with CaM-dependent targets

Enzyme	k_{off} (s^{-1})	k_{on} ($M^{-1}\ s^{-1}$)	K_d (calculated) $K_d = k_{off}/k_{on}$
Calcineurin	$(1.2 \pm .05) \times 10^{-3}$	4.6×10^7	28 pM
Calcineurin (a subunit only)	$(2.6 \pm 0.5) \times 10^{-4}$	ND	
αCaMKII	1.10[a]	2.1×10^{7}[b]	52 nM
αCaMKII + ATP	8.4×10^{-5}[a]	1.1×10^{7}[b]	7.6 pM
MLCK[c]	0.56	1.1×10^8	5.1 nM

Data come from [a]Putkey and Waxham [10], [b]Gaertner and Waxham, unpublished (but see also [17]), and [c]Torok and Trentham [18].

Perrino et al. [9] measured the dissociation rate of Ca^{2+}/CaM from CaN and determined two rates of ~ 0.4 and $4\ s^{-1}$. These rates were consistent between both CaNα and CaNβ isoforms of CaN. These dissociation rates are two orders of magnitude faster than those we measured. We are at a loss to explain these differences. Both studies used $CaM(C75)_{ACR}$ as a probe to measure the rate of dissociation, and the experimental conditions that can be identified are very similar. In attempts to resolve the difference, purified CaN was provided by Dr. Brian Perrino, and we found a similar slow dissociation rate in steady-state fluorimetry experiments as those reported above. In total, we measured this slow dissociation in five different preparations of CaN, and consistently arrived at a dissociation rate of $\sim 0.0013\ s^{-1}$. Therefore, we are confident in the values reported but cannot identify a reason for the differences.

CaN is a critically important enzyme that controls a number of fundamental cellular processes [4,15]. We have a particular interest in understanding the role of Ca^{2+}/CaM-signaling in the regulation of synaptic plasticity, and CaN has been directly implicated in the process of long-term depression [16]. Accurate knowledge of the kinetics of enzyme activation, which are in large part dictated by the binding kinetics of Ca^{2+}/CaM, is imperative for understanding CaN's role in plasticity. We have now established quantitative values for the rate of Ca^{2+}/CaM association and dissociation from CaN and the dissociation rate is significantly lower than those reported previously. Therefore, the values used in previous modeling studies to address CaN's role in various biological processes have significantly underestimated the tight binding of Ca^{2+}/CaM to CaN.

Acknowledgments

The authors thank Dr. Brian Perrino for reagents, many helpful discussions, and for suggestions on the manuscript. We also thank Dr. John Putkey for use of the stopped-flow instrument and for his critical reading of the manuscript. This work was supported by a grant from the NIH/NINDS NS50944. Joanna Forbes was supported in part by the University of Texas Medical School at Houston Summer Research Program.

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